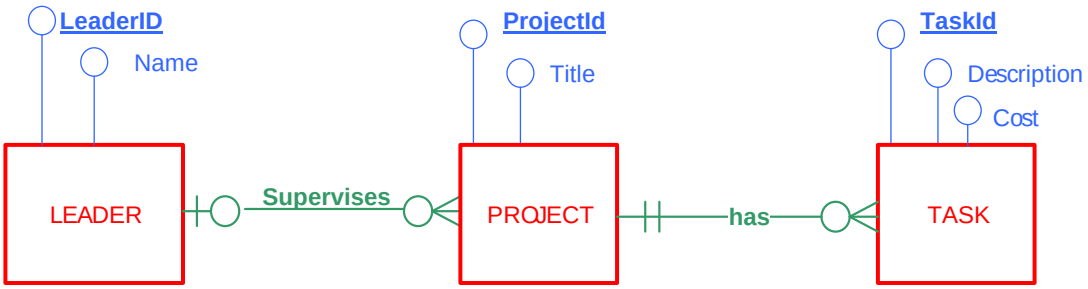
# INF10025 Data Management and Analytics

## Tutorial 6 – Discussion Questions for Week 6




**1.** Consider the ERD above. When converted to a Relational Schema:
   a. How many Foreign Keys will exist in the Task relation?
   b. How many Foreign Keys will exist in the Project relation?

**2.** Consider the Project Relation (after conversion to a relational schema).
   a. What is the name of the Foreign Key in this relation?
   b. Would this foreign key be allowed to have Null values?

**3.** Consider the Task Relation (after conversion to a relational schema).
   a. What is the name of the Foreign Key in this relation?
   b. Would this foreign key be allowed to have Null values?

**4.** Imagine that you now want to build the database based on the Relational Schema.
   a. Which table would you have to create first? Leader, Project or Task?
   b. Which table would you have to create second? Leader, Project or Task?
   c. Which table would you have to create last? Leader, Project or Task?

**5.** What is a Parent row in a relational database?

**6.** What is a Child row in a relational database?

**7.** True or False
   a. Leader is a parent of Project.
   b. Task is a parent of Project.
   c. Leader is a parent of Task.
   d. Task is a parent of Leader.

**8.** When using Oracle in default mode, nominate if the following statements are True or False
   a. A parent row that has no child rows may be deleted without error.
   b. A child row that has a parent row may be deleted without error
   c. A child row may not be orphaned
   d. A parent row cannot exist without having child rows
   e. A parent row that has one or more child rows may be deleted without error.

**9.** Consider the following data in the Person table. How many rows are selected by the following Queries?

| IdNo | Name | Gender | QtyA | QtyB |
|---|---|---|---|---|
| 23 | Johnny Storm | M | 10 | 10 |
| 46 | Eric Twinge | M | 7 | 4 |
| 17 | Diana Prince | F | 4 | 7 |
| 28 | Susan Storm | F | 1 | 3 |
| 134 | Dick Grayson | m | 6 | 2 |
| 5 | Barbara Gordon | F | 8 | 7 |
| 24 | Selina Kyle | F | 4 | 0 |
| 78 | Rick Hunt | M | | |
| 81 | Ted Howler | M | 6 | 6 |
| 89 | Sonita Marks | F | 3 | 6 |
| 2 | Dave Orkle | M | 2 | 4 |

**a.**
```
SELECT Name, QtyA+QtyB AS "Qty Total"
FROM person
WHERE QtyA+QtyB < 10;
```

**b.**
```
SELECT Name  FROM person
WHERE QtyA >=5
AND QtyB < 5;
```

**c.**
```
SELECT Name  FROM person
WHERE NOT (QtyA >=5 AND QtyB < 5);
```

**d.**
```
SELECT Name  FROM person
WHERE gender = 'F' OR gender = 'M'
AND QtyA+QtyB < 5;
```

**e.**
```
SELECT Name  FROM person
WHERE NOT (QtyA + QtyB)  < 10
AND Gender = 'F' ;
```

**f.**
```
SELECT  Name FROM person
WHERE  UPPER(Name) Like '%RI%'
OR   UPPER(Name) Like '%IN%' ;
```

**g.**
```
SELECT  Name FROM person
WHERE  LOWER(Name) Like '%or%';
```

**h.**
```
SELECT  Name FROM person
WHERE  UPPER(Name) Like '%Son%'
```

**i.**
```
SELECT  Name FROM person
WHERE  Gender = 'M'
OR         QtyA >= 5
AND       Gender = 'F'
OR         QtyB < 2;
```

**j.**
```
SELECT  Name FROM person
WHERE  QtyA > 2
AND       Gender = 'F'
OR         Gender = 'M';
```

**k.**
```
SELECT  Name FROM person
WHERE     IdNo > 0
AND        IdNo < 20
OR         QtyB < 5;
```

**l.**
```
SELECT  Name FROM person
WHERE  Gender = 'M' OR QtyA > 4
AND  Gender = 'F'  OR QtyB < 5;
```

**m.**
```
SELECT  Name FROM person
WHERE  Gender = 'M' AND Gender = 'F'
OR   QtyB >0   AND  QtyB < 5;
```

Assume that a relational schema and database has been built based on the ERD in Question 1.

LEADER(LeaderId, Name)

PK (LeaderId)

PROJECT(ProjectId, Title, LeaderId)

PK (ProjectId)

FK (LeaderId) References Leader

TASK (TaskId, Description, Cost, ProjectId)

PK (TaskId)

FK (ProjectId) References Project

**10.** Write a single SQL statement using **INNER JOIN** clauses and **aliases** that displays the **project title** and **leader name** for each of the projects in the project table:

**11.** Write a single SQL statement using **INNER JOIN** clauses and **aliases** that displays the **task description and cos**t, **project title** and **leader name** for each of the tasks in the task table.